

Xavier Rao

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Objective

Seeking a Software Engineering internship using strong programming skills in Python, Java and JavaScript. Eager to collaborate with project team members to help build cutting-edge solutions across diverse applications. Available January to August 2027.

Education

Rochester Institute of Technology (RIT)

Bachelor of Science, Software Engineering

Expected May 2028

Related Courses: Intro to Software Engineering, Web Engineering, Engineering of Software Subsystems, Software Design for Computing Systems, Intro Artificial Intelligence, Machine Learning, Web & Mobile I, Engineering Cloud Software Systems

Skills

Languages: Java, Python, JavaScript, Typescript, HTML/CSS, SQL, C, PHP, Terraform

Frameworks: Angular, React.js, Node.js, Flask, Spring Boot, Express.js

Software: Visual Studio Code, IntelliJ IDEA CE, STM32CubeIDE, Git, GDB, Maven, Jacoco, AWS

Projects

Most Likely To Game, *Personal Project*

May 2025 – April 2026

- Developed and deployed a real-time multiplayer web party game using React, JavaScript, Node.js, and Socket.IO, featuring an imposter mechanic where one player secretly receives a different AI-generated question each round
- Implemented WebSocket communication for live game state synchronization, integrated an AI API for dynamic question generation, and added features including scoring, spectator mode, and automatic reconnection handling

NutriApp, *Class Project*

Spring 2025

- Collaborated in a team to design and develop a multi-user nutritional planning application using Java, Angular, TypeScript, HTML, and CSS, featuring calorie tracking, meal planning, workout logging, and automated goal transitions
- Architected a modular backend applying GoF design patterns (Command, Composite, Strategy, State, Factory, Adapter) across subsystems for food tracking, shopping list generation, and data export in JSON, CSV, and XML formats

TypeSpeedrun, *Personal Project*

October 2024 – March 2025

- Developed a web-based typing speed assessment tool inspired by Monkeytype.com using HTML, CSS, and JavaScript
- Designed a dynamic mode selection interface for customizable tests (time, word count, quote length) with Fetch API integration for real-time JSON data retrieval

Cat Rescue, *Class Project*

Fall 2024

- Collaborated in a team of three to develop a web application for cat rescue in Rochester using Angular CLI, Java, HTML, CSS, and TypeScript
- Designed and integrated key pages (Login, Basket, Checkout, Admin) with Angular's HTTPClient for seamless frontend-backend communication

Work Experience

MLC STEM Center | Bloomfield, NJ

Summer 2022/2024/2025

STEM Mentor/Counselor

- Developed and delivered lesson plans on coding, physics, chemistry, and 3D design for classes of 20 students
- Designed and led a week-long curriculum introducing artificial intelligence concepts through hands-on coding projects
- Fostered collaborative learning through hands-on instruction, emphasizing foundational programming concepts

TAG Online Inc. | Clifton, NJ

Spring 2023

Intern

- Designed and developed custom websites using WordPress, HTML, CSS, and PHP, meeting client specifications for usability and aesthetics
- Contributed to projects including pinkhammerhome.com and corbocollection.com/hotels

NanoGraf | Chicago, IL

June 2021 – July 2021

Intern

- Engineered high-performance battery cells by formulating specialized slurries and applying precision compression in a controlled glove box environment
- Conducted patent research and competitive analysis, documenting findings in detailed spreadsheets to support materials innovation